Using Normalizing Flows for Weighting Estimators with Continuous Treatment

Chanhyuk Park Xiangyu Song

May 1, 2026

Abstract

1 Introduction

In observational studies, political scientists care about an unbiased estimator for average treatment effect (ATE) or average treatment effect on the treated (ATT). One way to generate an unbiased and also semiparametrically efficient estimator is by using augmented inverse propensity weighting(AIPW) estimator, known to be doubly robust. However, there are two caveats here. First, as continuous treatment is common in political science studies, simply dichotomizing treatments to fit into AIPW with a binary treatment framework may lose information on the treatment effect heterogeneity. Second, although researchers can use logit or probit to estimate the propensity score with binary treatment easily, an accurate estimation of the propensity score under continuous treatment is still a challenge. In this paper, we propose to use the normalizing flows (NF), which produce efficient and exact propensity score estimation with continuous treatment.

2 Related Literature

2.1 Continuous Treatment and Generalized Propensity Score

A large literature studies the inference for various treatment effects with binary or categorical treatments, either with randomized experiment or observational studies. In political science, such examples include the Get Out the Vote (GOTV) campaign on voter turnout (Gerber and Green, 2000), the effect of product differentiation in economic markets on firm-level lobbying in political markets (Kim, 2017), and the effect of patron-client network on economic development in China (Jiang, 2018). However, many empirical settings could also involve continuous treatment, such as the campaign spending, the intensity of a policy implementation, as well as the media exposure. In such settings, causal effects are naturally summarized by an average dose-response function, or the average structural function (Colangelo and Lee,

2026).

$$m(t) = \mathbb{E}\left[Y(t)\right],$$

rather than by a single average treatment effect. Under consistency, overlap, and conditional ignorability assumption, this curve can be identified as

$$m(t) = \mathbb{E}\left[\mu(X,t)\right], \qquad \mu(x,d) = \mathbb{E}\left[Y \mid X = x, T = t\right].$$

Hirano and Imbens (2004) provide the foundational propensity-score approach for continuous treatment: the generalized propensity score (GPS). Defined as

$$g(t \mid x) = f_{T|X}(t \mid x),$$

GPS is the conditional density of the treatment given the covariates. They have shown that the GPS plays an analogous balancing role to the binary propensity score. Although their framework provides a way to estimate does-response functions by adjusting for the GPS, it often relies on parametric modeling of the treatment density.

2.2 Semiparametric Theory and Doubly Robust Estimators

As mentioned by Kennedy et al. (2017), semiparametric theory is not directly available if we only assume mild smoothness conditions on $\theta(d)$ (e.g., differentiability) since without parametric assumptions $\theta(d)$ is not pathwise differentiable, and root n consistent estimators do not exist. By defining a parameter that represents the average outcome under an intervention that randomly assigns treatment based on the density of the continuous treatment, Kennedy et al. (2017) construct the efficient influence function for this stochastic-intervention parameter, which contains a pseudo-outcome term that attains the double robustness.

3 Problem Setup

4 Normalizing Flows

NF are generative models which produce tractable distributions where both sampling and density evaluation can be efficient and exact, using invertible functions. In other words, it is a technique that maps a random variable with a complex probability density function to the simpler density function.

A typical NF procedure starts with a simple base distribution such as Gaussian and then attempts to transform that function into a more complex distribution by applying a finite number of invertible functions[1]. The process utilized the *change of variables formula*, which guarantees mapping of one probability density function for a random variable Y with another probability density function of random variable Z:

$$\begin{aligned} p_{\mathbf{Y}}(\mathbf{y}) &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det \mathrm{D}\,\mathbf{f}(\mathbf{y})| \\ &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det \mathrm{D}\,\mathbf{g}(\mathbf{f}(\mathbf{y}))|^{-1}, \end{aligned}$$

where $\mathbf{g}$ is an invertible function, $\mathbf{f} = \mathbf{g}^{-1}$, and $\mathrm{D}\,\mathbf{f}(\mathbf{y}) = \frac{\partial \mathbf{f}}{\partial \mathbf{y}}$ is the Jacobian of $\mathbf{f}$. Figure 1 illustrates this relationship. Since $\mathbf{f}$ moves in the direction of a simple base distribution, it is called *normalizing flow*.

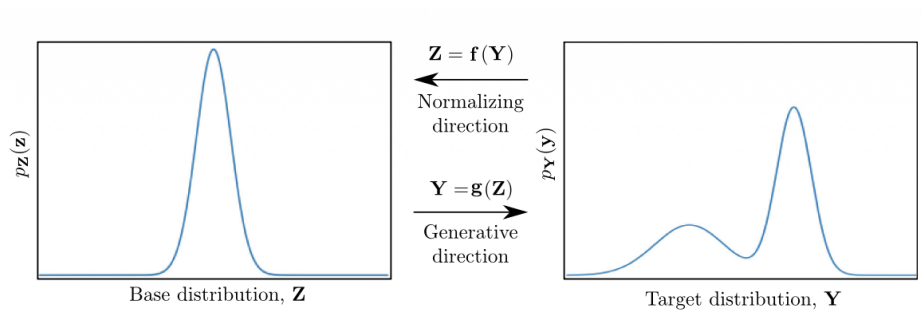

Figure 1: Graphical Illustration of Normalizing Flows. from Kobyzev et al. (2021)

[1]The existence of this process is always guaranteed, and discussed in Chen and Gopinath (2000).

As a result of this process, we can link the complex distribution (such as a joint distribution of variables we observe) with simple base distribution such as Gaussian. Since this is an invertible process, instead of dealing with the complex distribution itself, we can work with the simple base distribution such as sampling and density estimation, and then easily bring the result to the complex one.

NF is a useful tool in density estimation compare to other frequently used density estimation method. Histogram is a simple exploratory method. However, it relies heavily on the size of bins and thus often performs poorly on complex distributions. Unlike the histogram, KDE technique produces a smooth estimate of the density, using all sample points' locations. The main idea behind KDE has been proposed long ago (Rosenblatt, 1956), but the recent development of computing technology makes it a popular option today. KDE, however, requires researchers to decide which kernel to be used and the size of the bandwidth, and both influence the estimation result. Since NF usually relies on neural network, the computational cost may be high for complex design, it is more capable of dealing with complex density.

5 Using NF for Propensity Score Estimation

One straight forward application of density estimation with NF is propensity score estimation. Statistical estimators employing propensity score relies heavily on the accuracy of the propensity score.

Although there are criticism on using machine learning algorithm in propensity score estimation, it becomes more popular (Goller et al., 2020). Lechner and Okasa (2024) utillized random forest estimation of propensity score. In this paper, we propose to use NF as an alternative for propensity score estimation and leverage the estimated propensity score into the AIPW estimator in observational studies.

References

Chen, S. and R. Gopinath (2000): "Gaussianization," in *Advances in Neural Information Processing Systems*, ed. by T. Leen, T. Dietterich, and V. Tresp, MIT Press, vol. 13.

Colangelo, K. and Y.-Y. Lee (2026): "Double Debiased Machine Learning Nonparametric Inference with Continuous Treatments," *Journal of Business & Economic Statistics*, 44, 67–79.

Gerber, A. S. and D. P. Green (2000): "The effects of canvassing, telephone calls, and direct mail on voter turnout: A field experiment," *American political science review*, 94, 653–663.

Goller, D., M. Lechner, A. Moczall, and J. Wolff (2020): "Does the estimation of the propensity score by machine learning improve matching estimation? The case of Germany's programmes for long term unemployed," *Labour economics*, 65, 101855.

Hirano, K. and G. W. Imbens (2004): "The Propensity Score with Continuous Treatments," in *Applied Bayesian Modeling and Causal Inference from Incomplete-Data Perspectives*, John Wiley & Sons, Ltd, chap. 7, 73–84.

Jiang, J. (2018): "Making Bureaucracy Work: Patronage Networks, Performance Incentives, and Economic Development in China," *American Journal of Political Science*, 62, 982–999.

Kennedy, E. H., Z. Ma, M. D. McHugh, and D. S. Small (2017): "Non-Parametric Methods for Doubly Robust Estimation of Continuous Treatment Effects," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 79, 1229–1245.

Kim, I. S. (2017): "Political Cleavages within Industry: Firm-level Lobbying for Trade Liberalization," *American Political Science Review*, 111, 1–20.

Kobyzev, I., S. J. Prince, and M. A. Brubaker (2021): "Normalizing Flows: An Introduction and Review of Current Methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43, 3964–3979.

Lechner, M. and G. Okasa (2024): "Random Forest estimation of the ordered choice model," *Empirical Economics*.

Rosenblatt, M. (1956): "Remarks on Some Nonparametric Estimates of a Density Function," *The Annals of Mathematical Statistics*, 27, 832–837.